

Immunopathology Basics

William K. Dolen, MD • 49-slide current deck • slide companion + mechanism-first review

5 official objectives

Gell & Coombs I-IV

Type I deep dive

Mechanism → therapy

ScholarRx linked

SOURCE PRIORITY: the current 49-slide Dolen deck controls slide order, terminology, examples, and emphasis. The Word guide is used only where it adds useful explanation and is explicitly labeled Companion-only enrichment if that material is not in the current deck.

Official Learning Objectives

9.1 Compare IgE-mediated Type I hypersensitivity with Types II, III, and IV, and recognize clinical examples.

9.2 Relate allergy and atopy to clinical practice.

9.3 Describe the sequence leading to mast-cell activation, mediator release, and activation of other effector cells.

9.4 Recognize mediators released by mast cells, basophils, and eosinophils and distinguish early from late phase reactions.

9.5 Relate therapeutic interventions for allergic, autoimmune, and inflammatory diseases to the mechanisms driving disease.

Slides 2-6 and 26-27: The Four Mechanisms

Classic Gell & Coombs

Mechanisms of immunopathology (hypersensitivity)

- Type I: IgE mediated immediate hypersensitivity
 - rhinitis, asthma, atopic dermatitis, anaphylaxis
- Type II: IgG (IgM) mediated cytotoxic or stimulatory/inhibitory reactions
 - drug-induced hemolysis
- Type III: IgG (IgM) immune complex reactions
 - serum sickness; Arthus reaction
- Type IV: T cell mediated delayed hypersensitivity (Th1)
 - contact dermatitis, tuberculin sensitivity



Slide 3. The lecture's core classification. Type I is IgE/immediate; Type II is IgG/IgM against a cell or receptor; Type III is circulating immune complexes; Type IV is T-cell mediated delayed hypersensitivity.

Type	Primary mechanism in this deck	Deck examples	Recognition shortcut
I	IgE-mediated immediate hypersensitivity	Rhinitis, asthma, atopic dermatitis, anaphylaxis	IgE + FcεRI + mast-cell activation
II	IgG/IgM cytotoxic OR stimulatory/inhibitory	Drug-induced hemolysis; Rh disease; Graves; myasthenia gravis	Antibody binds a <i>fixed</i> cell-surface antigen/receptor
III	IgG/IgM immune-complex reaction	Serum sickness, Arthus reaction, hypersensitivity pneumonitis	Soluble Ag-Ab complexes travel and deposit
IV	T-cell mediated delayed hypersensitivity, usually Th1	Contact dermatitis, tuberculin sensitivity	No antibody required; 1-3 day delay

Clinical Caveat

- The mechanisms can (and do) **overlap!**
- In clinical practice, the pathophysiology of a given disease can involve more than one mechanism, even **simultaneously**
- A classic example is systemic lupus erythematosus
- Some agents (e.g., penicillin) cause one type of reaction in one person, and another type in a different person
- Exam questions will always ask for the predominant mechanism



Slide 4. Faculty caveat: mechanisms overlap in real patients. For exam-style questions, identify the predominant mechanism being tested.

FACULTY WORDING WORTH REMEMBERING

Diseases can use more than one mechanism simultaneously. SLE is the lecture's classic overlap example, and even the same trigger such as penicillin can cause different hypersensitivity mechanisms in different people.

Companion-only mnemonic: ACID = **A**naphylactic/Allergic I, **C**ytotoxic II, **I**mmune complex III, **D**elayed IV. Useful memory aid, but the current deck itself does not use this mnemonic.

Slides 7-13: Allergy, Atopy, and Semantic Traps

What Is Allergy?

- 'Allergy' is a very specific disease **mechanism**: IgE-dependent Type I immediate hypersensitivity, not a disease and not a diagnosis
- This **mechanism** can trigger symptoms of many diseases (rhinitis, asthma, atopic dermatitis, etc.)
- For those diseases, symptoms can also be triggered by **other mechanisms** (like ILCs)
- To avoid semantic confusion, be careful how you use the term 'allergy' with patients, in writeups, and with colleagues



Slide 10. The deck's preferred clinical definition: 'allergy' is an IgE-dependent Type I mechanism, not a disease or diagnosis.

NEED TO MEMORIZE

Allergy = mechanism, not diagnosis. In this lecture's preferred US clinical usage, allergy means IgE-dependent Type I immediate hypersensitivity. A disease such as asthma or rhinitis can still be triggered by non-IgE mechanisms.

Vocabulary trap: different specialties and the lay public use the word “allergy” differently. The lecture explicitly warns you to use the term carefully in patient communication and writeups.

Atopic / “allergic” diseases

- The classic short list in the deck is **rhinitis, asthma, and atopic dermatitis**.
- These tend to occur together in individuals and families, described in the lecture as the **atopic march**.
- Other diseases can involve IgE or other mechanisms: urticaria, anaphylaxis, drug/venom reactions, eosinophilic gastroenteritis.

The Allergic Diseases



- Some others include urticaria, anaphylaxis, drug reactions, and insect venom reactions, and eosinophilic gastroenteritis. Not inherited within families.
- Symptoms can be triggered by mechanisms other than Type I immediate hypersensitivity (allergy)
- One important mechanism involves the epithelial alarmin **TSLP** (thymic stromal lymphopoietin), which can activate ILC2 cells that produce cytokines traditionally associated with Th2 lymphocytes
- The TSLP-ILC2 pathway is so new that it has no place in the Gell and Coombs classification
- Recall that the therapeutic monoclonal antibody tezepelumab binds TSLP



Slide 13. A major deck-only nuance: TSLP from epithelium can activate ILC2 cells and generate Th2-like cytokine biology without IgE. Tezepelumab targets TSLP.

Deck-only high-yield connection: the **TSLP** → **ILC2** pathway sits outside classic Gell & Coombs. This is why “allergic disease” symptoms do not automatically prove an IgE mechanism. The treatment connection appears again on slide 46 with **tezepelumab**.

Slides 14-25: Type I Hypersensitivity in Depth

Step 1 - Prime the effector cell

Pulling the Trigger

- The **FcεRI** high-affinity IgE receptor is found on:
 - Mast cells
 - Basophils
 - Activated eosinophils
 - Some other cells
- It binds IgE
- The receptor-IgE complex waits for allergen exposure
- Cross-linking of IgE causes receptors to aggregate and activate
- The end result is sneeze, wheeze, or worse

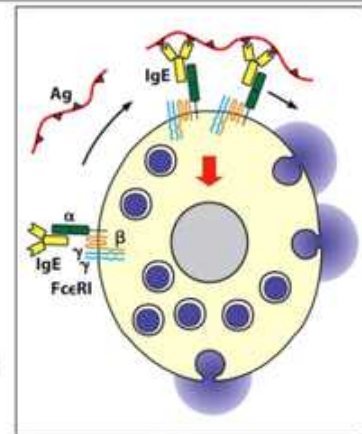


Figure 13.3 The Immune System, 4th ed. © Garland Science 2009



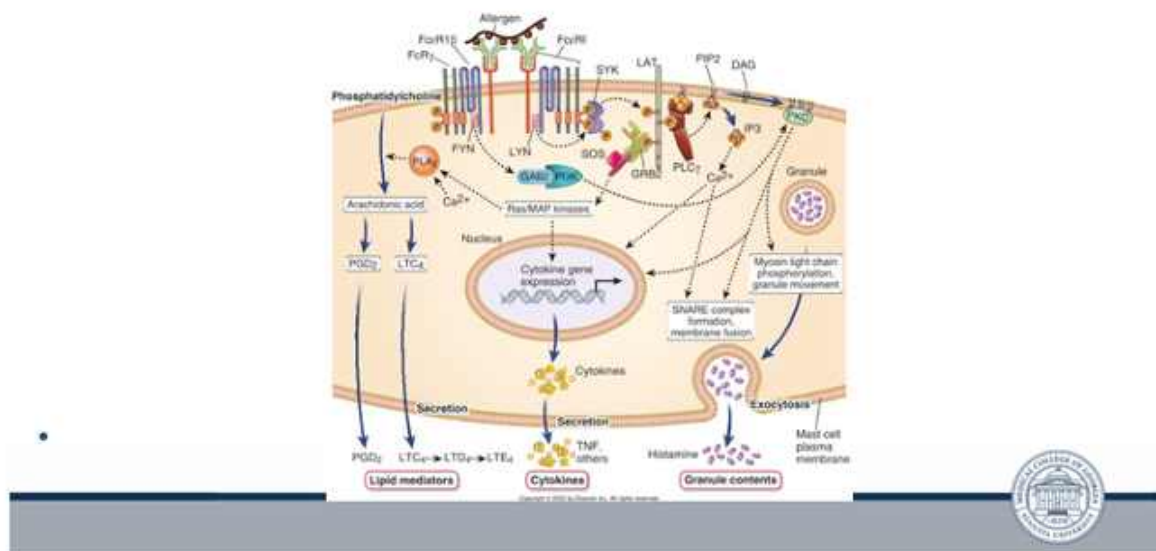
Slide 15. FcεRI is the high-affinity IgE receptor on mast cells, basophils, activated eosinophils, and some other cells. IgE waits on the receptor until allergen cross-linking triggers activation.

MECHANISM CHAIN

IgE bound to FcεRI → allergen cross-links IgE → receptor aggregation → mast-cell activation → mediator release → immediate symptoms + inflammatory recruitment.

Step 2 - Intracellular signaling and mediator release

BTK in Mast Cells



Slide 17. The BTK/mast-cell signaling figure links receptor cross-linking to kinase signaling, calcium-dependent exocytosis, lipid mediator synthesis, and cytokine production.

WHAT TO TAKE FROM THE BTK FIGURE

You do not need to turn the entire pathway into a kinase list. The conceptual output is three parallel products: **granule contents** (including histamine), **new lipid mediators**, and **cytokines**. The treatment slide later mentions **-brutinibs**, connecting therapy back to signaling.

Mast cell vs basophil

- Both are major Type I effectors, express FcεRI, can release histamine, and can also be activated by non-IgE mechanisms.
- **Tryptase is a clinically useful marker of mast-cell activation.**
- The deck explicitly says **basophils do NOT have tryptase.**

Eosinophil products

Class of product	Product	Biological effects	
Enzyme	Eosinophil peroxidase	Toxic to parasites and mammalian cells by catalyzing halogenation Triggers histamine release from mast cells	Preformed in eosinophil
	Eosinophil collagenase	Remodeling of connective tissue matrix	
Toxic protein	Major basic protein	Toxic to parasites and mammalian cells Triggers histamine release from mast cells	
	Eosinophil cationic protein	Toxic to parasites Neurotoxin	
	Eosinophil-derived neurotoxin	Neurotoxin	
Cytokine	IL-3, IL-5, GM-CSF	Amplify eosinophil production by bone marrow Cause eosinophil activation	Synthesized after eosinophil activation
Chemokine	CXCL8	Promotes influx of leukocytes	
Lipid mediator	Leukotrienes C ₄ , D ₄ , and E ₄	Cause smooth muscle contraction Increase vascular permeability Cause mucus secretion	
	Platelet-activating factor	Chemotactic to leukocytes Amplifies production of lipid mediators Activates neutrophils, eosinophils, and platelets	

Figure 12.9 The Immune System, 3rd, (© Garland Science 2009)



Slide 19. The eosinophil table separates preformed granule products from mediators synthesized after activation.

Preformed eosinophil products	Major effect shown in deck
Eosinophil peroxidase	Toxic to parasites/mammalian cells; can trigger mast-cell histamine release
Eosinophil collagenase	Connective-tissue matrix remodeling
Major basic protein	Toxic to parasites/mammalian cells; can trigger mast-cell histamine release
Eosinophil cationic protein	Toxic to parasites; neurotoxic
Eosinophil-derived neurotoxin	Neurotoxic

Synthesized after activation	Deck effect
IL-3, IL-5, GM-CSF	Amplify eosinophil production and activation
CXCL8	Promotes leukocyte influx
Leukotrienes C4, D4, E4	Smooth-muscle contraction, ↑ vascular permeability, mucus secretion
Platelet-activating factor	Chemotaxis, amplifies lipid mediators, activates neutrophils/eosinophils/platelets

Early vs late phase

Early vs, Late Phase Type I Reactions

- With allergen challenge, an “early phase” reaction occurs within minutes, due to release of pre-formed mast cell mediators
- A “late phase” reaction can occur several hours later due to an inflammatory response
- Don’t get the “late phase reaction” (Type I) confused with delayed-type hypersensitivity (Type IV)

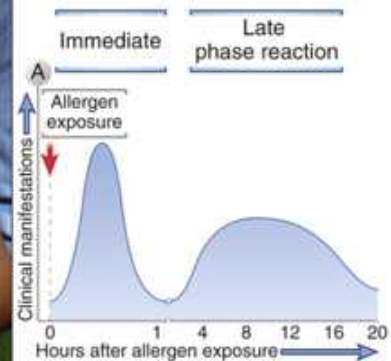


Figure 14.27 The Immune System, 6th ed. © Garland Science 2015



Slide 22. Early phase occurs within minutes from preformed mast-cell mediators; late phase occurs hours later from the inflammatory response.

Ozzy checkpoint - click him once you can explain why a Type I *late phase* reaction is still not Type IV.

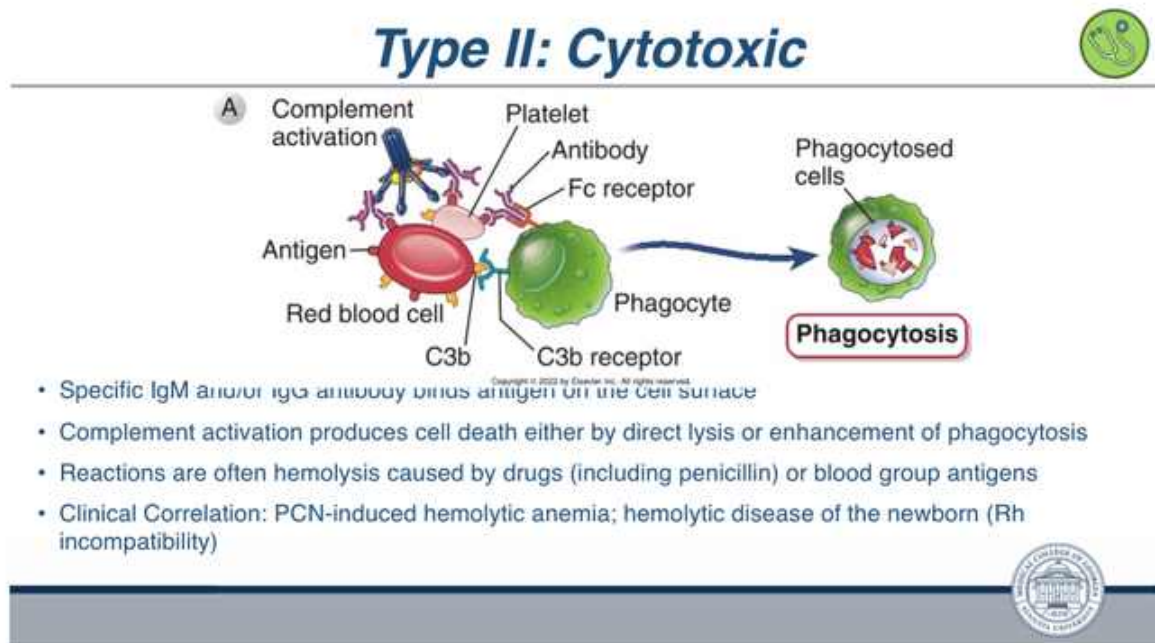
Classic trap: late-phase Type I happens several hours after allergen exposure, but it remains part of an IgE/mast-cell-centered Type I response. **Type IV delayed hypersensitivity** is T-cell mediated and classically appears 1-3 days after antigen contact.

Selected disease factoids from slides 23-25

- Rhinitis/asthma: IgE mechanisms are common, but ILC pathways can also contribute.
- Atopic dermatitis: lecture phrase is the “**itch that rashes**”; IgE may trigger itching, scratching produces rash. “Eczema” is a physical finding, not one diagnosis.
- Chronic urticaria is **very rarely IgE-dependent**; autoantibody to FcεRI can drive mast-cell activation.
- Anaphylactic vs anaphylactoid terminology: either mechanism can be mild, severe, or fatal.

Companion-only enrichment: the Word guide adds sensitization details (Th2, IL-4/IL-13 class switching, IL-5 eosinophils), skin-prick/specific-IgE testing, and IM epinephrine physiology for anaphylaxis. These are useful clinically, but they are *not* shown as dedicated content in the current 49-slide deck, so they are not treated as faculty slide emphasis here.

Slides 27-30: Type II - Antibody Against a Fixed Target



Slide 28. Type II cytotoxic mechanism: IgG/IgM binds a cell-surface antigen; complement can cause direct lysis or enhance phagocytosis.

TYPE II RECOGNITION RULE

The antigen is fixed to the cell/tissue/receptor. Ask: "What exactly is the antibody binding?" If the answer is a surface antigen or receptor, Type II rises quickly.

Cytotoxic

Penicillin-induced hemolytic anemia;
hemolytic disease of the newborn/Rh
incompatibility.

Stimulatory / inhibitory

Graves: stimulate receptor. Autoimmune
hypothyroidism: block TSH receptor.
Myasthenia gravis: block ACh receptor.

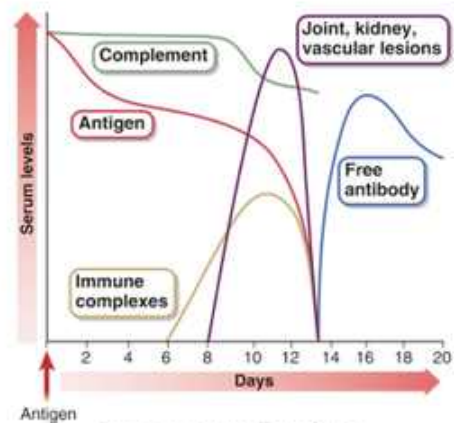
SLIDE 29 NUANCE

The deck notes that stimulatory/inhibitory Type II reactions were once colloquially called "Type V." For this course, keep them under **Type II**.

Slides 31-39: Type III - Immune Complex Deposition

Immune Complexes

- Range in size from one protein bound to one Ab to millions of components
- Size regulated by relative Ag and Ab concentrations
 - Big IC get bound by complement and cleared from the circulation - no problem
 - Medium IC trapped in small blood vessels will activate complement in situ, causing vasculitis or mast cell activation
 - Smallest IC pass into tissues, degraded - no problem



Slide 32. Immune-complex size depends on antigen/antibody proportions. Medium complexes can become trapped in small vessels and activate complement in situ.

TYPE III RECOGNITION RULE

Soluble antigen + antibody → circulating immune complexes → deposition → complement/inflammation.

Where complexes become trapped	Clinical pattern named on slide 33
Glomerulus	Nephritis
Pleura	Pleuritis
Peritoneum	Peritonitis
Synovium	Arthritis
Choroid plexus	Cerebritis
Postcapillary venules	Vasculitis

Type III Reactions



1. Arthus reactions - local in tissue
 - The year 2026 is the 123rd anniversary of the Arthus reaction - described by Nicholas Maurice Arthus
2. Serum sickness - systemic
3. Hypersensitivity pneumonitis - inhaled



Slide 35. The deck divides Type III into three recognizable patterns: local Arthus reaction, systemic serum sickness, and inhaled-antigen hypersensitivity pneumonitis.

Arthus

Local tissue reaction in a sensitized person with preformed IgG; local IC deposition and complement activation. Deck example: sore arm a few hours after dT booster.

Serum sickness

Large foreign antigen exposure; rash, glomerulonephritis, arthritis, fever as antibody develops, usually about a week.

Hypersensitivity pneumonitis

Inhaled antigen → IgG response → IC deposition in alveoli. Deck examples: farmer's lung, bird fancier's lung.

TYPE II VS III DISCRIMINATOR

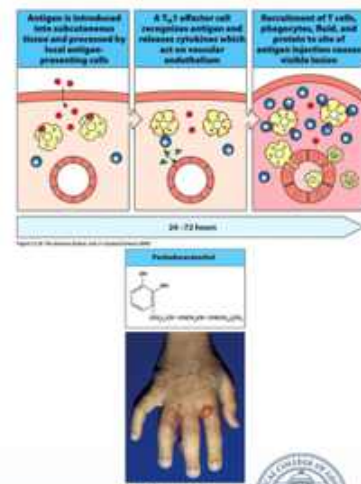
Type II = antibody binds a fixed target.

Type III = soluble complexes form first, then deposit elsewhere.

Slides 40-44: Type IV - T-Cell Mediated

Type IV – Inflammatory Th1

- Th1 response mediated by effector T cells, macrophages
- Processed Ag peptides presented by MHC Class II
- Some Ag are haptens, e.g. nickel, PCN, pentadecacatechol

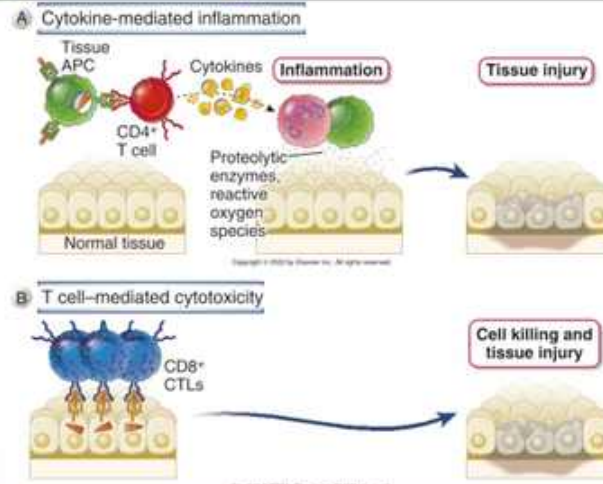


Slide 41. The inflammatory Th1 pathway: processed antigen is presented by MHC II; effector T cells and macrophages drive the response. Haptens such as nickel can trigger it.

TIMING + MECHANISM

Type IV = T cells, not antibody, and typically 1-3 days after antigen contact.

Type IV – Cytokine or Tc cell



Slide 43. The deck broadens Type IV beyond Th1 cytokine inflammation to include cytotoxic T-cell mechanisms that directly kill target cells.

- **DTH** = delayed-type hypersensitivity when the response is pathologic/detrimental.
- **CMI** = cell-mediated immunity when the same kind of T-cell biology is beneficial against pathogens.
- Other examples in the deck: transplant rejection, resistance to fungal infection, Hashimoto thyroiditis.
- **Clinical anergy** = inability to mount Type IV responses; slide 44 gives sarcoidosis as an example of general anergy.

Slides 45-47: Treat the Mechanism

Therapeutic Options



- Corticosteroids, other immunosuppressants (jakinibs, -brutinibs)
- Specific monoclonal antibodies (e.g., omalizumab, benralizumab, tezepilumab)
- Mediator-specific agents
 - H1 antihistamines
 - Leukotriene receptor antagonists
- Allergy shots (only for IgE-proven rhinitis, asthma, selected other conditions)



Slide 46. The lecture's therapy slide: broad immunosuppression, signaling inhibitors, monoclonal antibodies, mediator-specific agents, and allergen immunotherapy.

Therapeutic strategy	Examples explicitly on the deck	Mechanistic logic
Broad / pathway immunosuppression	Corticosteroids; JAK inhibitors; -brutinibs	Reduce inflammatory signaling rather than one end mediator
Specific monoclonal antibodies	Omalizumab, benralizumab, tezepelumab	Target an immune driver/effector pathway
Mediator-specific agents	H1 antihistamines; leukotriene-receptor antagonists	Block downstream products of Type I activation
Allergy shots	For IgE-proven rhinitis, asthma, selected other conditions	Mechanism-specific immunotherapy rather than nonspecific symptom labeling

LECTURE'S FINAL PRINCIPLE

Clinically identifying the mechanism guides management. The point of learning I-IV is not just categorization; it lets you predict what part of the immune response is rational to target.

Companion-only enrichment: the Word guide additionally names methotrexate/DMARDs, anti-TNF and anti-IL-17 biologics, and describes allergen immunotherapy as promoting tolerance/IgG4. Those details are helpful but are not on the current deck's therapy slide, so memorize the deck's own treatment list first.

High-Yield Traps & Vignette Recognition

If the vignette says...	Think...	Why
Immediate sneeze/whoeze, allergen, IgE, mast cell	Type I	FcεRI cross-linking and mediator release
Hemolysis after drug; antibody on RBC	Type II cytotoxic	Fixed cell-surface target
Graves or myasthenia	Type II receptor dysfunction	Antibody stimulates or inhibits a receptor
Rash + arthritis + nephritis after large antigen exposure	Type III serum sickness	Systemic circulating immune complexes
Local reaction hours after injection in sensitized patient	Type III Arthus	Local immune-complex deposition
Farmer/bird exposure with alveolar immune-complex disease	Type III HP	Inhaled antigen → IgG complexes
Nickel/contact reaction 1-3 days later	Type IV	T-cell/Th1 delayed response

ONE-LINE FINAL MAP

I = IgE immediate. II = antibody on fixed target. III = soluble complexes deposit. IV = T cells delayed.

Source hierarchy: Dolen 49-slide current deck is primary. The Word guide is secondary enrichment and does not control slide numbering or faculty emphasis.